

Figure 5 · TinyDCS three-layer architecture

### Layer 1 — Published mechanistic models

ADRAC  
(log-logistic AFT)

Conkin RM/NM  
(Eq. 14/15)

Gerth 3RUT-MBe1  
(bubble ODE)  
[calibration WIP]

### Layer 2 — TinyDCS surrogate (this paper)

LightGBM · 13 features · Smithson-Verkuilen target · logit transform  
ZERO-INFLATED 2-STAGE:  $P(y=0)$  classifier × continuous regressor on  $y>0$   
Split-conformal on logit residuals · Mahalanobis OOD abstention

### Layer 3 — Personalization + fusion (Paper 2)

Hierarchical Bayesian  
per-subject log  $\lambda$   
(conjugate Gaussian)

Wearable multimodal fusion  
HR, HRV, SpO<sub>2</sub>, skin T  
(scaffolding only)